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PHYSICS

★★ Important Topics and Formulass ★★

CLASS 10

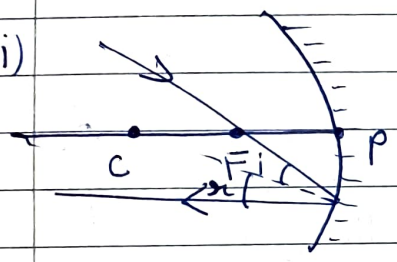
NCERT

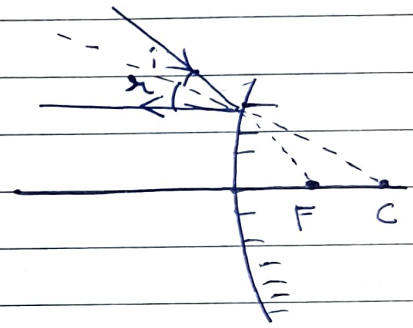
1. Laws of Reflection

2. Spherical Terms & terms: Pole, Focus, centre of curvature etc.

3. Image formation

(i)





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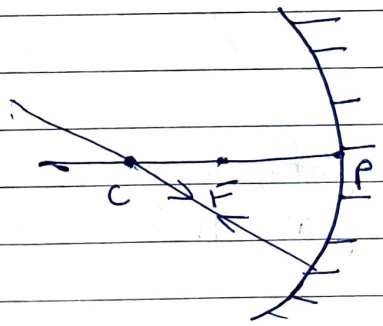

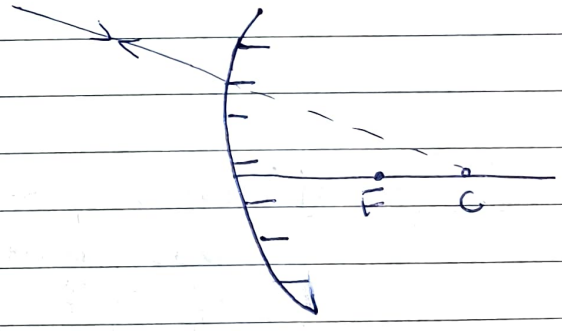

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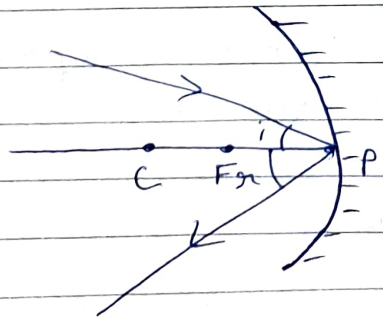

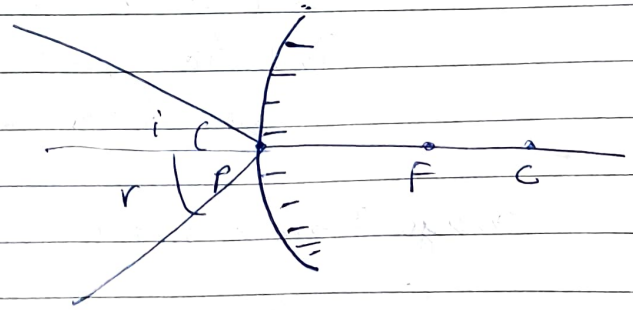

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TERMS

① Centre of reflecting surface of a spherical mirror is a pt. => Pole

② The reflecting surface forms a part of sphere whose center is known as centre of curvature.

③ The radius of sphere of which mirror forms a part is called radius of curvature.

④ St. line passing through the pole & centre of curvature is principal axis.

⑤ The pt. at which ∥ rays meet/intersect on the principal axis —> FOCUS

USES OF (i) concave mirror

- (i) torches, search lights, vehicles headlights to get ∥ beams of light
- (ii) shaving mirrors
- (iii) used by dentists
- (iv) to conc. sunlight to produce heat in solar furnaces

Q. State the conventions of New Cartesian Sign convention

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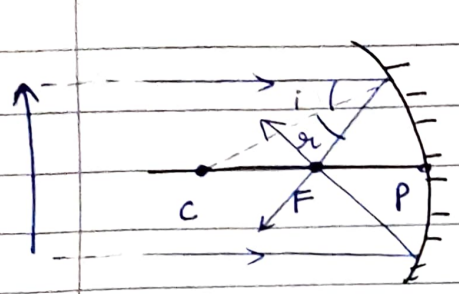
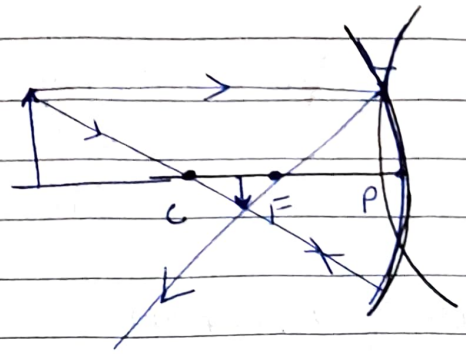
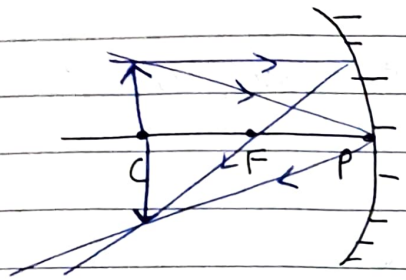
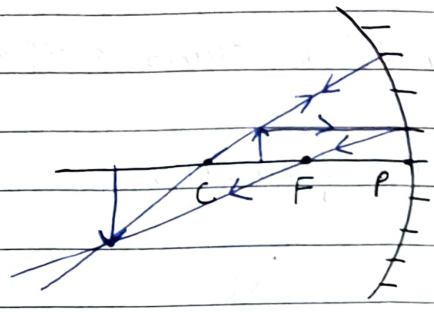
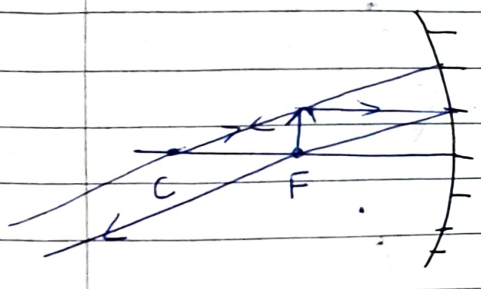
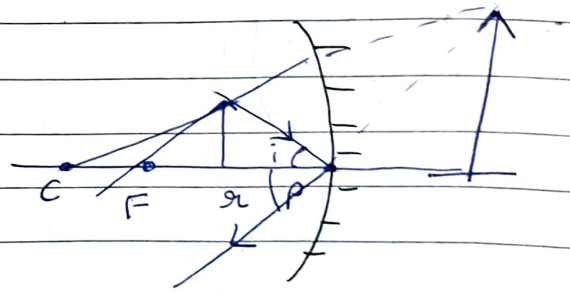












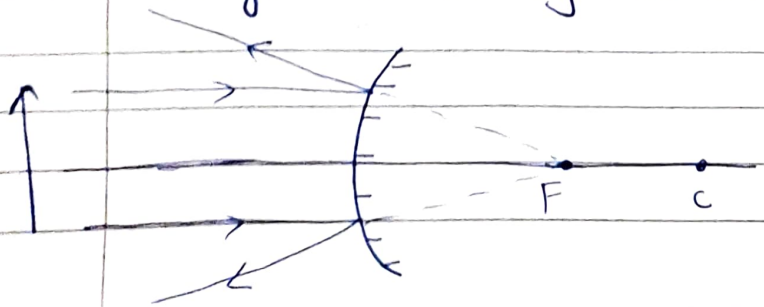
5. <u>Uses</u>: (i) torches, search-lights, vehicles headlights to get ∥ beams of light

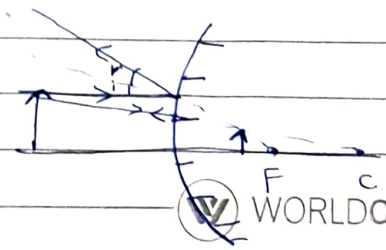
(ii) shaving mirrors

(iii) dentists

(iv) to conc. sunlight → to produce heat in solar furnaces

6. Image Formation by convex





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★ found in wall of Agra Fort facing Taj Mahal

7. <u>Uses</u> :- rear view (wing) mirrors
↳ have wides field of view as they are curved outwards

8. Sign Convention :- Rules [9.2.3]
~~Ru~~

9. $$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

10. Magnification :- gives the relative extent to which the image of an object is magnified w.r.t object size

$$\boxed{m = \frac{h'}{h} = -\frac{v}{u}}$$

11. <u>Refraction</u> : When light travels from one medium to another, its direction of propogation changes. This phenomena is known as refraction.

$\boxed{\text{CAUSE:}}$ Due to change in speed of light as it enters from one transparent medium to another

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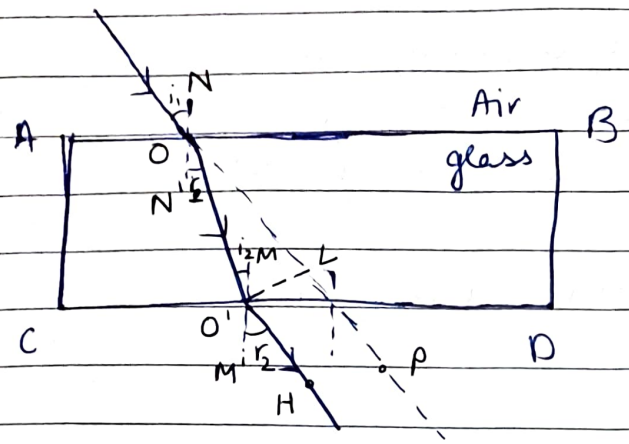
12. Refraction through a glass slab:-



Rarer → denser: towards N
denser → rarer: Away from Normal

Emergent ray is ∥ to direction of incident ray: extent of bending of ray of light is at the opp. ∥ faces is equal and opposite.

13. Laws of Refraction:-

(i) I.R, R.R & the N to the interface of two transparent media at the pt. of incidence, all lie on same plane.

(ii) ration of sine of ∠i to the sine of ∠refraction is a const. for the light of a given colour & for the given pair of media. (true for angle $0 < i < 90$)

$$\frac{\sin i}{\sin r} = \underbrace{\text{const}}$$

refractive index of 2nd medium w.r.t 1st

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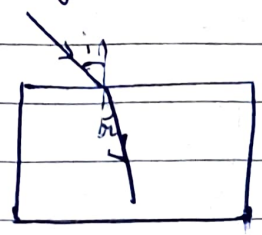
14. Refractive Index



↳ refractive index of med 2 w.r.t med 1 = n_{21}

$$n_{21} = \frac{\text{speed of light in med 1}}{\text{speed of light in med 2}} = \frac{v_1}{v_2}$$

also,

$$n_m = \frac{c}{v}$$

Optically denser medium may not possess greater mass density. eg. kerosene has higher ~~RI~~ N, so optically denser than water, although its mass density is less than water.

15. Refraction by spherical lenses

<u>Lens</u> :- lens is a transparent material bound by 2 surfaces

Thin lenses → small <u>apertures</u>

↓

Aperture: effective diameter of circular outline

→ centre of curvature equidistant from optical center

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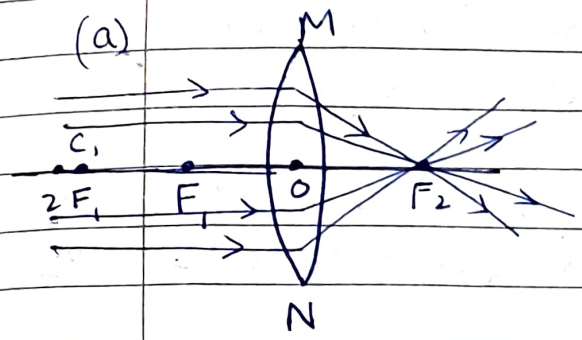
16. IMAGE FORMATION

(a)



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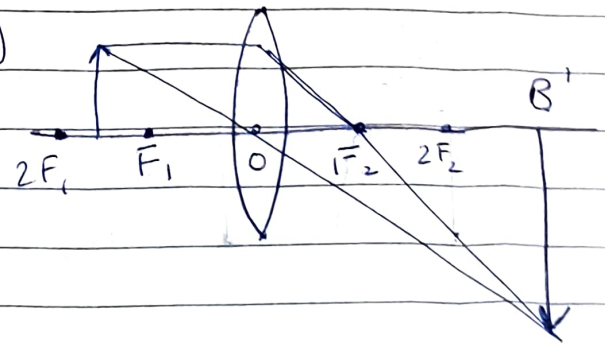

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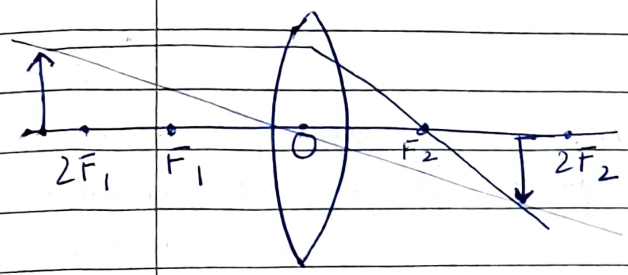

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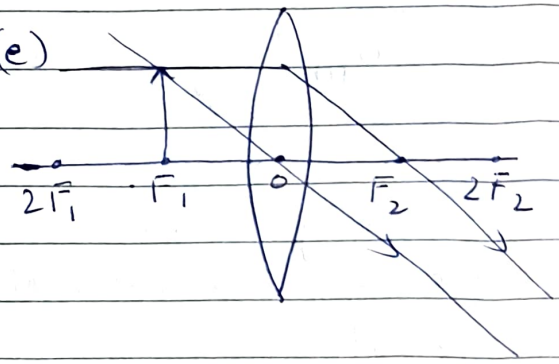

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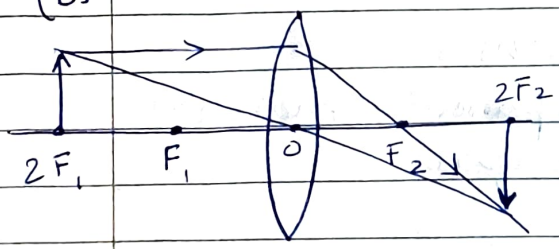

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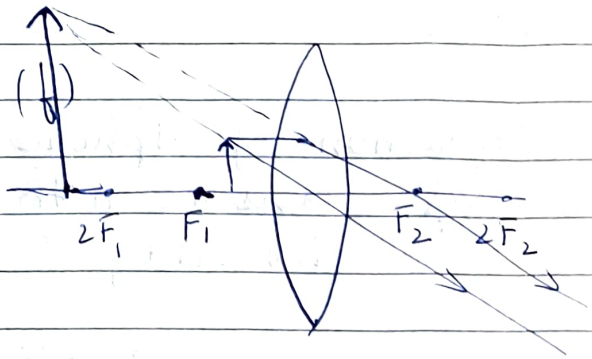

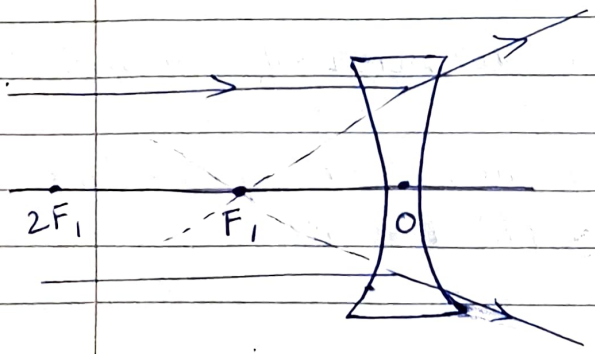

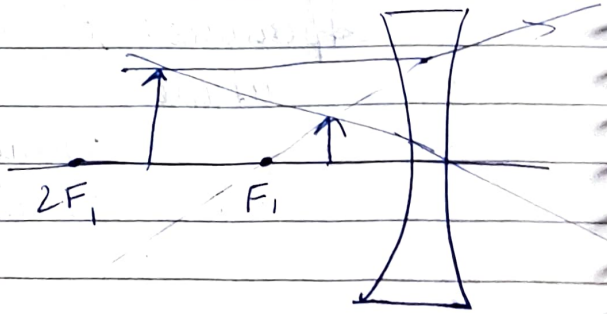

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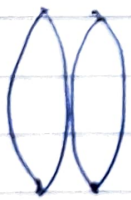
17. $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

18. <u>Power of lens:-</u> Degree of convergence or divergence of light rays achieved by a lens => Power

$P = \frac{1}{f}$

Diopter



$\rightarrow P = P_1 + P_2 + P_3 + \ldots$

(used in camera, microscopes, television)